

Randomized and low-rank approximation

Random column subsets of arbitrary fixed-sparsity matrices

Difficulty: challenging **Importance:** interesting to the community

Status: Solved

Rating rationale: Historical ratings for the original open target. Challenging because arbitrary support intersections defeat existing sparse-matrix estimates; community impact is understanding limits of sparse sketches and column selection.

Resolution: affirmative, 12 September 2026

Author: Sidney Holden, Center for Computational Biology, Flatiron Institute, Simons Foundation, New York, USA. [Official affiliation](#) verified 12 September 2026.

[Theorem 1.1 and Corollary 1.2](#) of *Random column subsets of fixed-sparsity matrices: an unrestricted theorem* prove the full statement below. The finite exponential tail bound covers arbitrary signs, repeated columns and unrestricted support intersections. Setting the aspect-ratio lower bound to $1/(2C)$ yields the exact asymptotic target. No growing-sparsity rate is claimed.

A separate [independent Codex AI-agent audit](#) returned PASS without mathematical corrections, satisfying the **Solved** policy. AI assistance was used for review and submission preparation; this is informal automated review, not human peer review or formal verification. No Lean verification was performed. [Proof source](#) · [Submission record](#).

Huang, Rudelson and Tikhomirov retain conjecture and prior-result credit. The original target and ID remain unchanged; the earlier checks below are historical.

Problem statement

Fix an integer $s \geq 2$ and $C \geq 1$. Suppose $r_k \leq k$, $k/r_k \rightarrow C$, and $n_k/r_k \rightarrow \infty$. For every deterministic sequence $M_k \in \{-1, 0, 1\}^{k \times n_k}$ with exactly s nonzero entries per column, and a uniformly random r_k -element column set I_k , prove or disprove

$$\forall \eta > 0, \quad \Pr\{\sigma_{\min}((M_k)_{:I_k}) > \eta\} \rightarrow 0.$$

Here $\sigma_{\min}(B) = \inf_{\|x\|_2=1} \|Bx\|_2$. There is no hypothesis controlling intersections of column supports.

Reference

Han Huang, Mark Rudelson, and Konstantin Tikhomirov, [Well-Invertible Column Subsets of Sparse Matrices Are Rare](#), §7, Conjecture 7.1; compare Theorem 1.3 for the additional structural hypothesis in the proved result.

Status check

Searches included “*unrestricted deterministic sparse sketches*” conjecture and the exact paper title with 2026. No subsequent resolution was located. The paper’s disproof of a particular SparseStack conjecture does not establish its own unrestricted deterministic conjecture.

Audit — 2026-09-10

Rechecked [Huang–Rudelson–Tikhomirov, Theorem 1.3 and Conjecture 7.1](#): a structural subclass is proved, while arbitrary deterministic supports remain conjectural. Title and sparse-subset follow-up searches found no full resolution. [Tikhomirov’s later embedding paper](#) studies random models and spectral-norm estimates, not all deterministic support patterns.